Adam Ingah

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Summary

U.S. citizen and MSEE candidate at UW (embedded systems & FPGA digital design) seeking hardware development, test, or embedded engineering roles. Hands-on experience building, testing, and debugging embedded hardware and electronics prototyping.

Education

University of Washington, Seattle | M.S. Electrical Engineering | 3.88 GPA | exp. *2027*
- Relevant Coursework: FPGA Digital Design, Embedded & Real-Time Systems, Software Engineering for Embedded Applications, TinyML, Analog Circuits for Sensor Systems

University of Alabama, Birmingham | B.S. Electrical Engineering | 3.6 GPA | May 2025
- Relevant Coursework: Signals & Systems, Communication Systems, Embedded Microcontrollers, Digital Logic Design
- High Distinguished Honors, Cum Laude

Experience

Mechanical Engineer | Amazon Leo | 9/2026 - Present
- Built, maintained, and managed a 3D printing farm to support print needs that enable the the Lab infrastructure team to deliver venues to customers
- Worked on 3D assemblies and designed mechanisms for custom fixtures/parts for ground station equipment and flat satellite builds

3D Printer Lab Assistant | UAB Libraries | 5/2022 - 5/2025
- 3D printer lab based in the library where I maintained/troubleshooted/repaired dozens of 3D printers using 3DprinterOS to supervise the prints as well as slice and validate the prints.
- Maintained inventory to keep track of filaments and material usage.
- Assisted Medical, Engineering, and other students with their CAD files and print projects

Prototype Design Lab | UAB School of Engineering | 8/2023 - 4/2024
- Worked with the Alabama Department of rehabilitation services to create an open source and affordable Low Vision Magnifier using off the shelf parts and 3D printed parts.
- Delivered a sub $200 alternative to $2000 low vision magnifiers saving costs by 90%

Projects

Autonomous Fire-Extinguishing Robot | Capstone Project | adamingah.com/projects/amfr
- Collaborated on a cross-functional capstone team to design, integrate, and debug an embedded system on Raspberry Pi (embedded Linux), combining thermal imaging and motor control for autonomous fire detection and suppression.
- Led hardware integration and board-level troubleshooting across the electrical/software boundary, verifying reliable sensor-actuator communication under test conditions.
- Prototyped and 3D-printed mechanical components in Fusion 360, iterating to meet form, fit, and durability requirements under test.

IoT Plant Irrigation System | adamingah.com/projects/ipws
- Programmed an ESP32 to stream soil-moisture sensor data over Wi-Fi to Arduino IoT Cloud, enabling real-time remote monitoring and control.

Skills

Programming:	C/C++, Python, SystemVerilog, Assembly, MATLAB, Docker, Git
Hardware:	FPGA, ESP32, Oscilloscopes, Multimeters, I2C/SPI/UART/GPIOs
Cad:	KiCad, LTspice, SolidWorks, Fusion 360, 3DPrinterOS
Tools:	Linux, System Testing & Validation, Requirements Documentation, PDS